higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1 It is a system used in process instruments in the design and configuration of automated and manual systems in various process plants. (2)

1.2 1.2.1
- Chemical industry
- Petrochemical industry
- Power generation
- Mineral processing industry
- Pulp and paper industry
- Water treatment
- Food and beverage industry (Any 3 × 2) (6)

1.2.2 Chemical industry
- These companies convert a number of natural resources,✓ which include water, metals and gasses,✓ into a wide variety of chemical products.✓

Petrochemical industry
- These are companies that manufacture chemical products✓ obtained from petroleum and other fossil fuels✓ that contain the same chemical compounds.✓

Power generation
- It produces and delivers electrical power.✓✓
- The generated power is delivered to areas through a power grid.✓

Mineral processing industry
- It is responsible for the mining or removal of minerals from soil deposits through underground excavations, open workings or processing plants.

Pulp and paper industry
- The fibres of wood, some fibrous crops and waste paper can be separated chemically through chemical processes, or mechanically, to form pulp.
- Mixing this pulp with water forms a suspension most commonly used to manufacture paper.

Water treatment
- Can be described as a mechanism and processes we use to treat water after it has been used in homes, businesses or industries (Any 3 × 3) (9)

1.3 1.3.1 Risky or dangerous

1.3.2 Able to burn, catch fire or ignite

(2 × 2) (4)
[21]

QUESTION 2

2.1 2.1.1 Used in differential pressure application with pressure gauges, pressure switches and pressure transmitters. (2)

2.1.2 Perfect sealing (1)

2.1.3 Butterfly valve (1)

2.1.4
- Greater force required for flow control
- Prone to cavitations (2)

2.1.5 Ball and cage valve (1)

2.1.6
- High capacity
- Low leakage
- Low maintenance requirements
- Tight sealing with low torque
- Long operating life
- Very high rangeability
- Good resistance to vibration and cavitation (3)

2.1.7 Relatively high cost (1)

2.1.8 Used where flow splitting is required or vice versa, especially in chemical and pharmaceutical industries. (2)

2.1.9
- Very large diaphragms may require great force.
- Limited sizes (2)

2.2 2.2.1
- Positioners
- Limit switches
- Fail-safe system

2.2.2 Positioners
- Positioners work well on valve actuators with springs. However they are vital to actuators without springs.✓
- They can also use positioners with accumulators.✓
- Positioners can supply air pressure to an actuator from the accumulator. This will let a valve open, close or fail.✓

Limit switches
- They can be used in motion-control, robotic-control and computer numerical control system.✓

Fail safe system
- It has many applications. Each application depends on the safety requirements of the system and the situation that must be avoided.✓ (Any 3 × 1)

2.2.3 Positioner

- The purpose is to adjust the actuator✓
- They are used with actuators to help them position valve correctly if there is a change in an air signal from a controller.✓
- In this way, the correct position in line with the input signal is maintained.✓

Limit switches

- They determine the actual position of process elements or equipment. With valves, limit switches will signal 'on' or 'off' depending on the position of the valve.✓
- They usually indicate whether the valve is fully shut or fully open.✓
- In this way, they make sure that the valve is in proper position and is not damaged.✓

Fail safe system

- To ensure that equipment fails in a safer manner.✓✓✓

(3 × 3) (9)

2.3
- Valve body
- Seat
- Disc
- Stuffing box
- Packaging gland
- Gland nuts

(Any 4 × 1) (4)

2.4
- Valve body

The fluid flows through this section✓

- Seat

The section of the valve that does not move✓

- Disc

Component that controls the fluid✓

- Stuffing box

Holds the packaging in place around the stem of the valve✓

- Packaging gland

Used to compress the packaging against the stem of the valve to ensure a good seal✓

- Gland nuts

Used to adjust the pressure on the packaging gland.✓ (Any 4 × 1) (4)

[32]

QUESTION 3

3.1 3.1.1 5 psi to bar
1 PSI = 0,0689 bar ✓
Therefore
5 psi = 5 × 0,0689✓
5 psi = 0,344 bar✓✓

3.1.2 15 psi to kPa
1 psi = 6,895 kPa✓
Therefore
15 psi = 6,895 × 15✓
15 psi = 103,425 kPa✓✓

(2 × 4) (8)

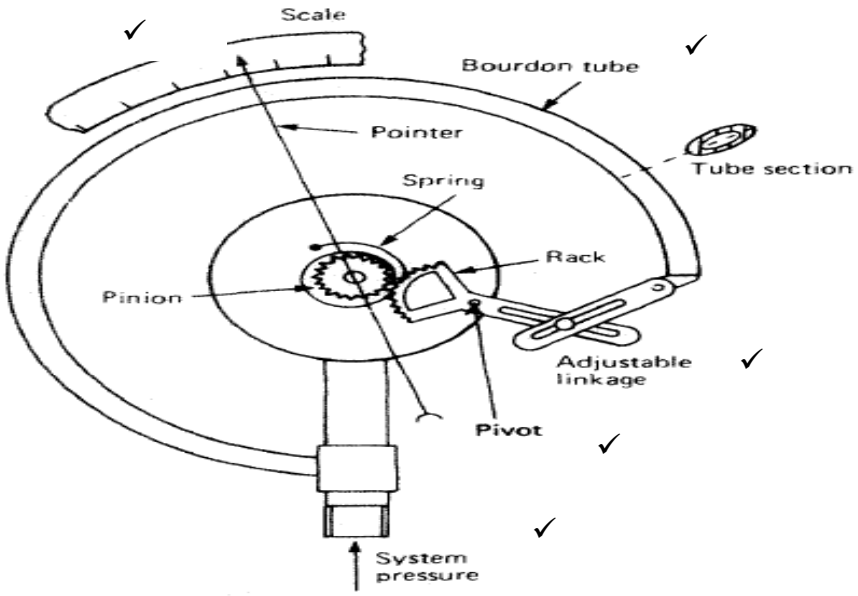
3.2 $\Delta P = p\ g\ h$✓
$h = \Delta P / p\ g$✓
$h = 200 \times 10^3 / 13,6 \times 10^3 \times 9,8$✓
h = 1,5 meter✓✓ (5)

3.3 3.3.1
- Steel
- Bronze
- Brass
- Stainless steel (Any 2 × 1) (2)

3.3.2
- Temperature
- Total tube length
- Wall thickness of a tube
- Material of a tube
- Radium of bend (5)

3.3.3
- Zero error
- Span error
- Angularity
- Hysteresis (Any 3 × 1) (3)

3.3.4



Sketch with labels = 7 marks (7)
[30]

QUESTION 4

4.1
4.1.1 False
4.1.2 True
4.1.3 True
4.1.4 False
4.1.5 True
4.1.6 True
4.1.7 True
4.1.8 False
4.1.9 False
4.1.10 False

(10 × 1) (10)

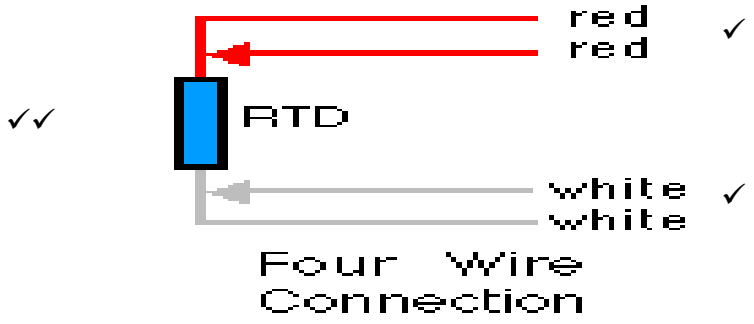
4.2 4.2.1 In 1821 Seebeck discovered that if you join two wires of different materials there will be an EMF generated in the circuit✓ if one junction is heated and the other is kept at a constant temperature.✓ The EMF and the current it generates are found to change in value due to the change in temperature between the hot and cold junctions.✓ Because the EMF is a function of the change in temperature it can be used as a measure of temperature of the hot junction if the other junction is kept at a known temperature, which is known as a reference temperature, if it is compensated for.✓

4.2.2 In 1824 Peltier discovered that if two different metal junctions are formed and if current is made to flow over the junctions then heat will be given off if current flows in one direction and absorbed if it flows in the opposite direction.✓ The amount of heat that is generated or absorbed is a function of the current flow.✓ The heating effect that takes place at the junction whenever current flows in one direction is an indication of the work that is done to overcome the junction EMF✓✓

4.2.3 It was first believed that the EMFs that are present at the junction of a thermocouple alone are an indication of Peltier effect and therefore the junctions are an only source of EMF in a thermocouple. Professor Thompson found in a series of experiments that if a wire has temperature gradient between its points✓ and if a current is made to flow in the wire then heat will be given off if current flows in one direction and absorbed if it flows in the other direction.✓ This means that an EMF is generated next to a heated wire from the hot junction to the cold side.✓

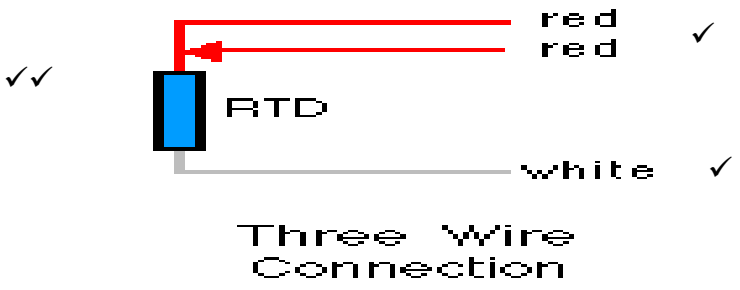
(3 × 4) (12)

4.3 4.3.1



2 marks for the wire connection
1 mark for the RTD
1 mark for the correct drawing (4)

4.3.2



2 marks for the wire connection
1 mark for the RTD
1 mark for the correct drawing (4)
[30]

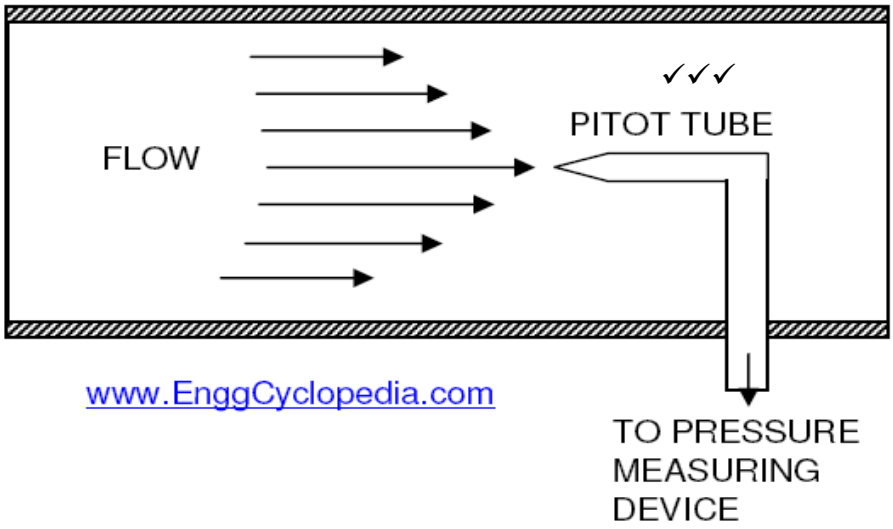
QUESTION 5

5.1

5.1.1	K
5.1.2	A
5.1.3	F
5.1.4	L
5.1.5	B
5.1.6	D
5.1.7	C
5.1.8	E
5.1.9	H
5.1.10	G

(10 × 1) (10)

5.2 5.2.1



Sketch with Labels = 5 marks (5)

5.2.2
- High pressure loss
- It cannot be used with liquids containing solid particles as these will form a layer of dirt on the plate which will thus cause an incorrect reading.
- With the area ratio greater than 0,55 it becomes impractical to use the orifice plate for accurate measurements. (3)

5.2.3
- Low manufacturing costs
- No maintenance
- Can be accurately machined
- Easy to install and replace (Any 3 × 1) (3)

5.2.4 If a tube is placed in a stream and the flow has a velocity V1 at a pressure P1,✓as the fluid approaches the body the fluid particles are decelerated until the velocity V2 is zero✓ at this point the pressure increases to P2.✓ If the pressure difference (P2–P1) can be measured, the velocity of the fluid can be found.✓ (4)

5.3 After you have decided where you are going to place the orifice plate flowmeter, you can install it by following these steps:✓

- Make sure the piping is not under pressure and that it is smooth and clean.✓
- Fit bolts in the pipeline flanges to keep the meter in its place.✓
- Put gaskets on either side of the body of the flowmeter and align them with the orifice plate.✓
- Certain orifice plate flowmeters have centralising tools, which fit between the pipeline flanges. Make sure that the meter is in the correct position when you do this.✓
- The meter is then tightened with bolts, while the centralising tool is rotated to push the bolts against the outer edge of the bolt holes.✓
- You can now fit the remaining bolts and tighten them according to the manufacturer's specifications or the relevant standards.✓ (7)

5.4 In order to zero check an orifice plate flowmeter, you can usually check whether the indicator indicates zero✓ when there is no flow.✓ Certain orifice plate flowmeters will allow you to make adjustments, for example, turning a screw✓ to ensure that the indicator shows zero.✓✓ (5)

[37]

TOTAL: 150

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